

Name: Eman Kashif

Course: Cloud Computing

Department: Software Engineering

Registration Number: FA24B1-SE-073

Section: A

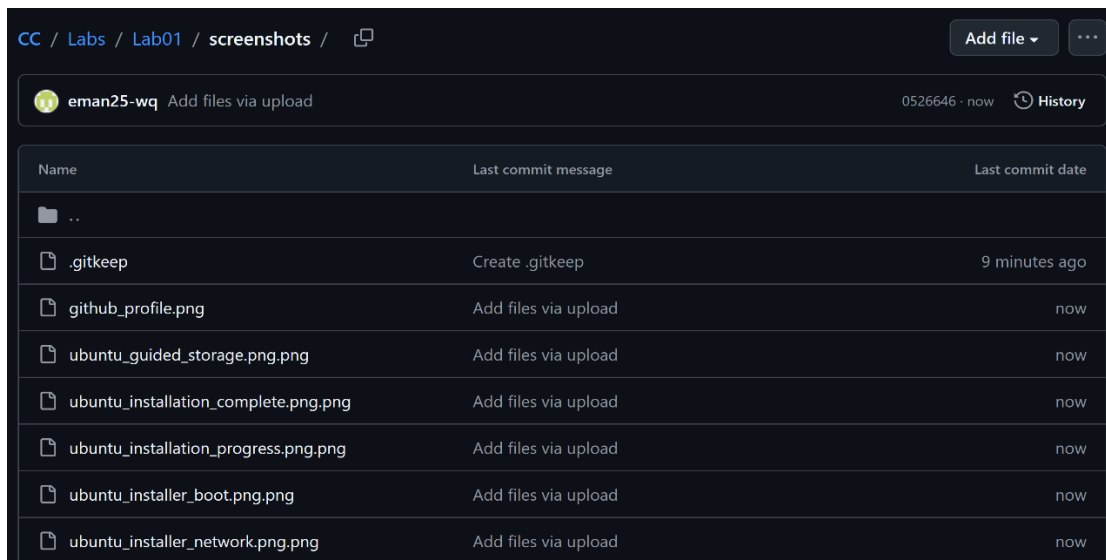
GitHub username: eman25-wq

Lab 01: GitHub Identity, Student Portal Registration, and Ubuntu Server Installation

Lab Objective: In this lab, you will create and correctly configure your GitHub account, register yourself in the Student Marks Portal, install Ubuntu Server in VMware Workstation Pro, verify your required Linux identity, and connect to the server remotely from Windows.

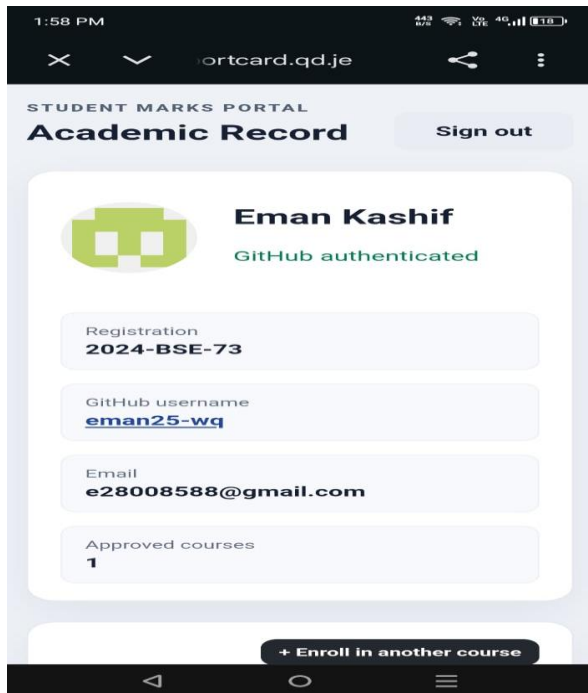
GitHub & Student Portal

1. lab01_folder_structure.png — Shows the required CC/Labs/Lab01 folder structure.



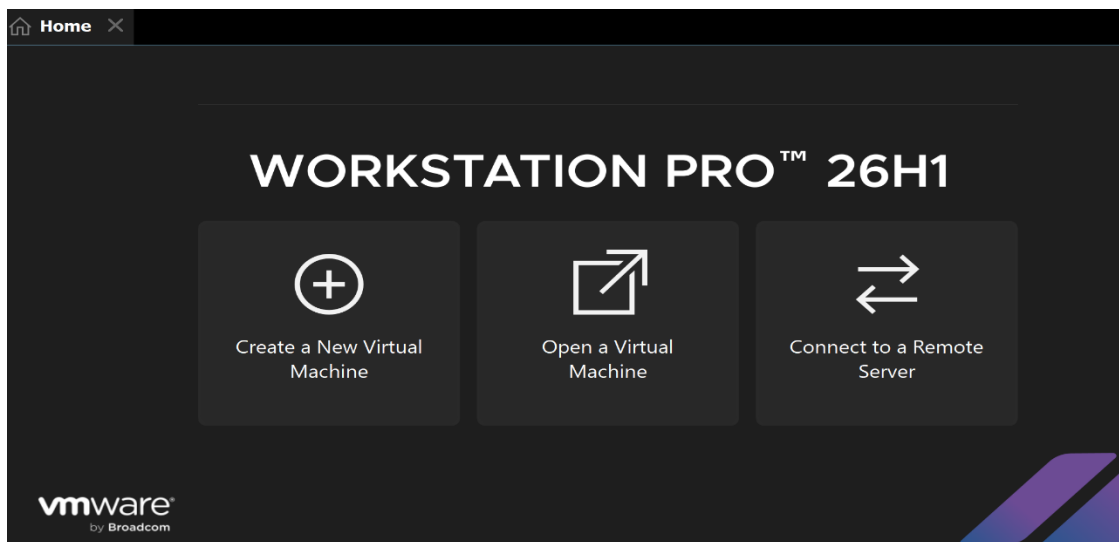
Name	Last commit message	Last commit date
..		
.gitkeep	Create .gitkeep	9 minutes ago
github_profile.png	Add files via upload	now
ubuntu_guided_storage.png.png	Add files via upload	now
ubuntu_installation_complete.png.png	Add files via upload	now
ubuntu_installation_progress.png.png	Add files via upload	now
ubuntu_installer_boot.png.png	Add files via upload	now
ubuntu_installer_network.png.png	Add files via upload	now

2. github_profile.png — Shows the GitHub profile configured with the student's identity.

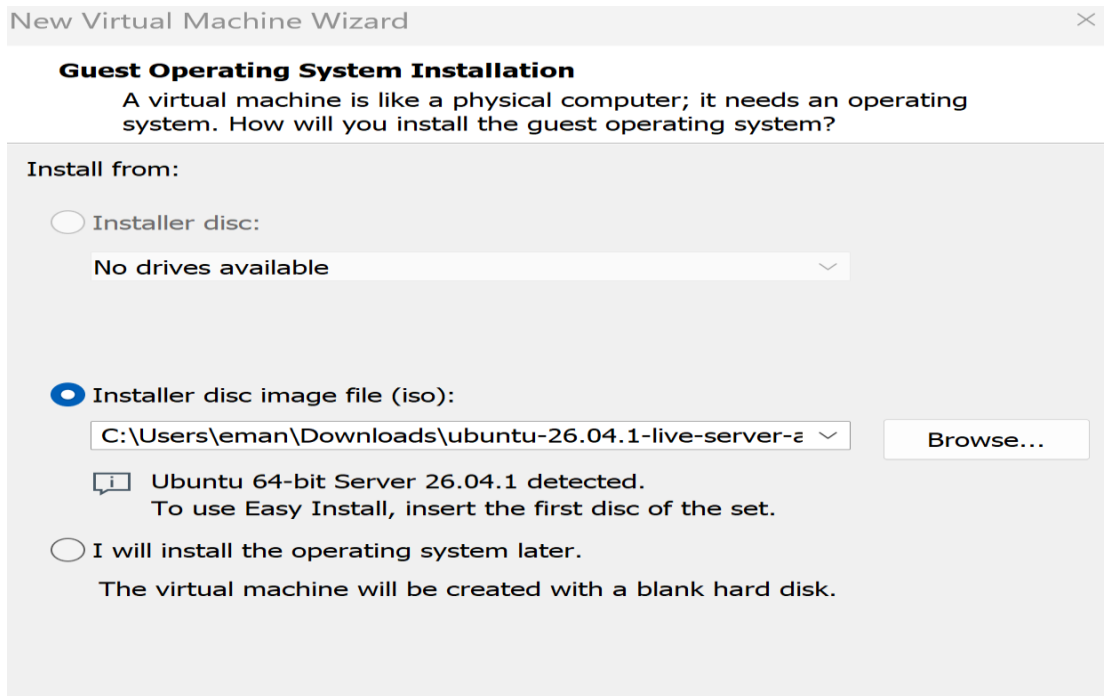


VMware & VM Creation

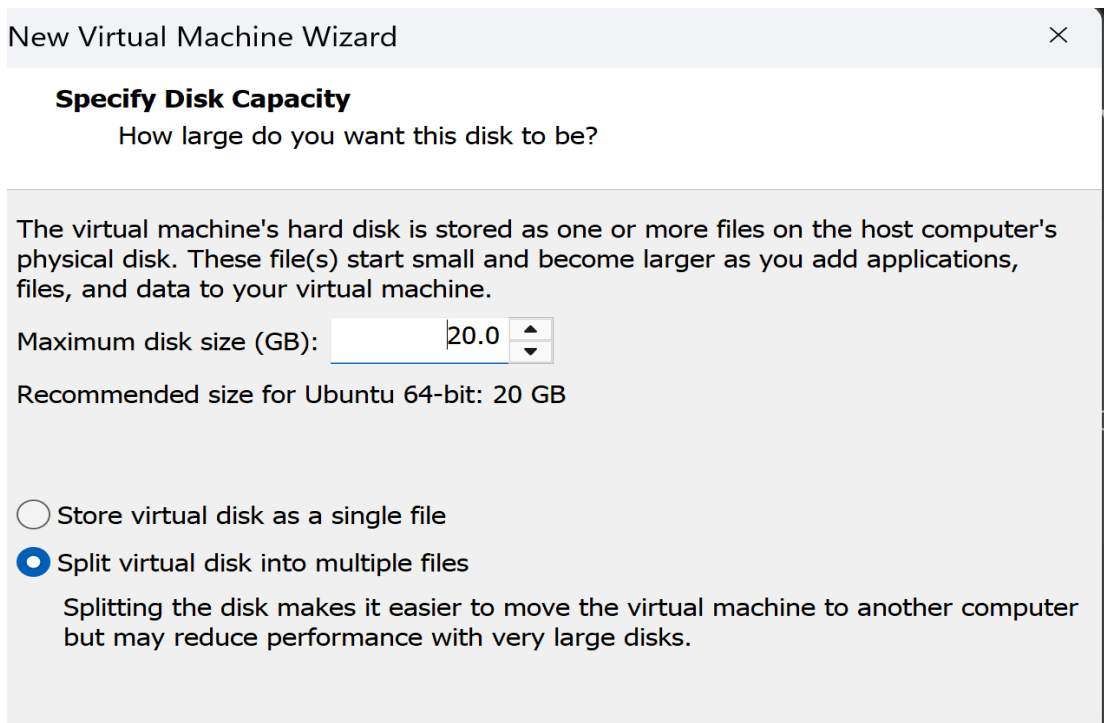
3. vmware_workstation.png — Shows VMware Workstation Pro installed and ready to create the virtual machine.



4. ubuntu_server_iso.png — Shows the downloaded Ubuntu Server ISO.



5. available_storage.png — Shows the available storage for the virtual machine.



6. vm_creation_wizard.png — Shows the VMware New Virtual Machine Wizard.

New Virtual Machine Wizard

Name the Virtual Machine

What name would you like to use for this virtual machine?

Virtual machine name:

Ubuntu-Server-Lab01

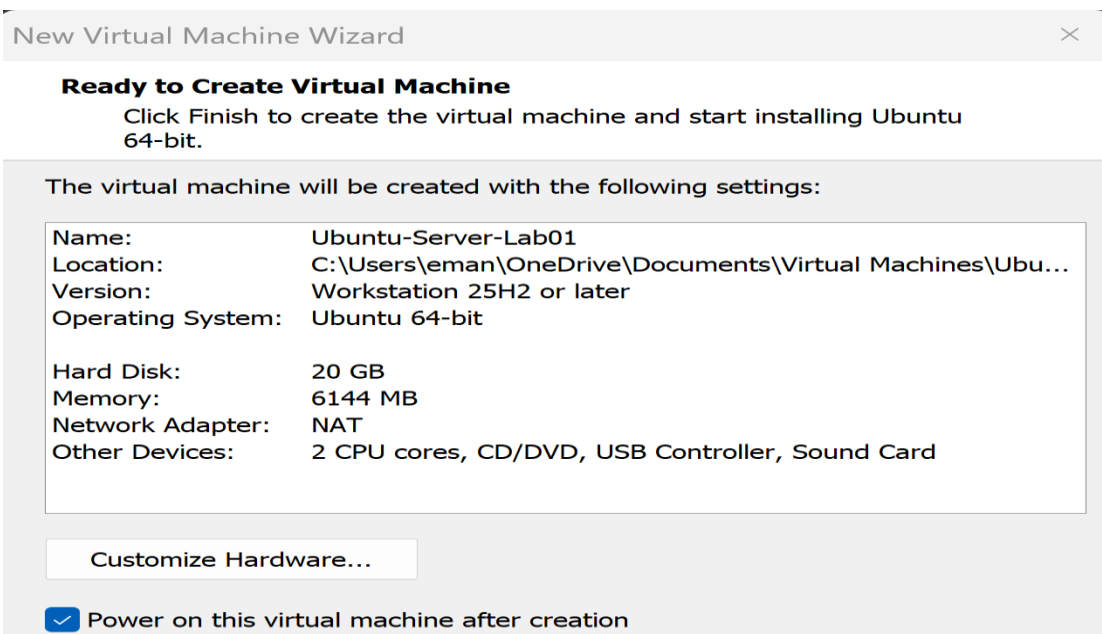
Location:

C:\Users\eman\OneDrive\Documents\Virtual Machines\Ubuntu-Se

Browse...

The default location can be changed at Edit > Preferences.

7. `vm_configuration_summary.png` — Shows the final VM configuration before creation.



Ubuntu Installation

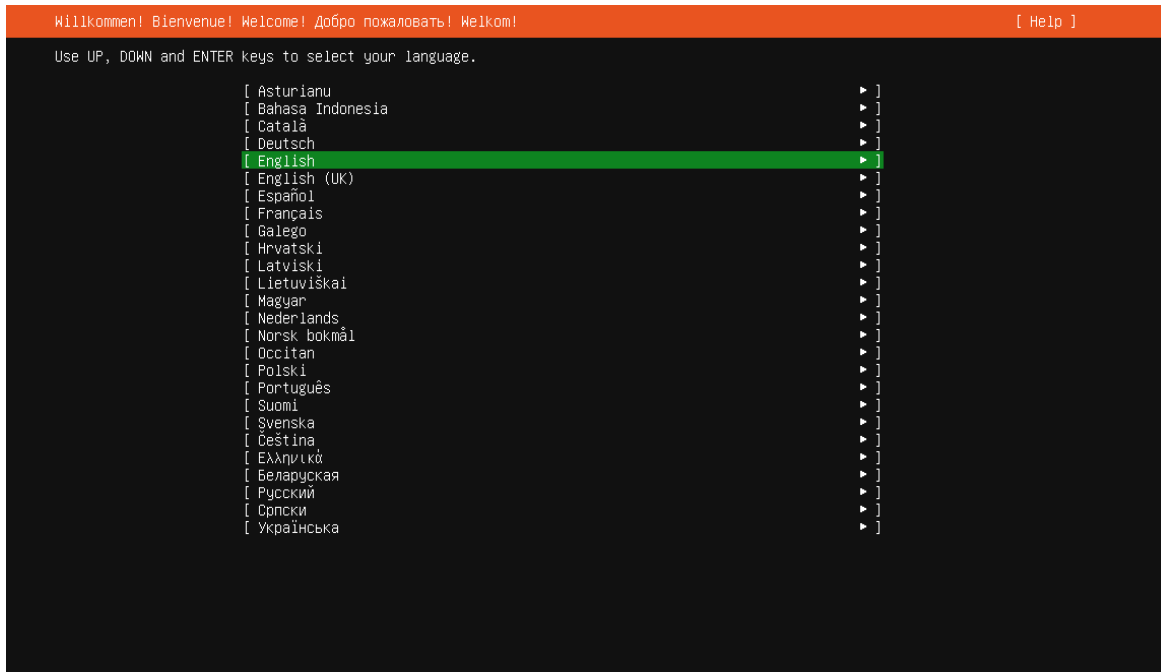
8. `ubuntu_installer_boot.png` — Shows the Ubuntu Server installer boot screen.

```

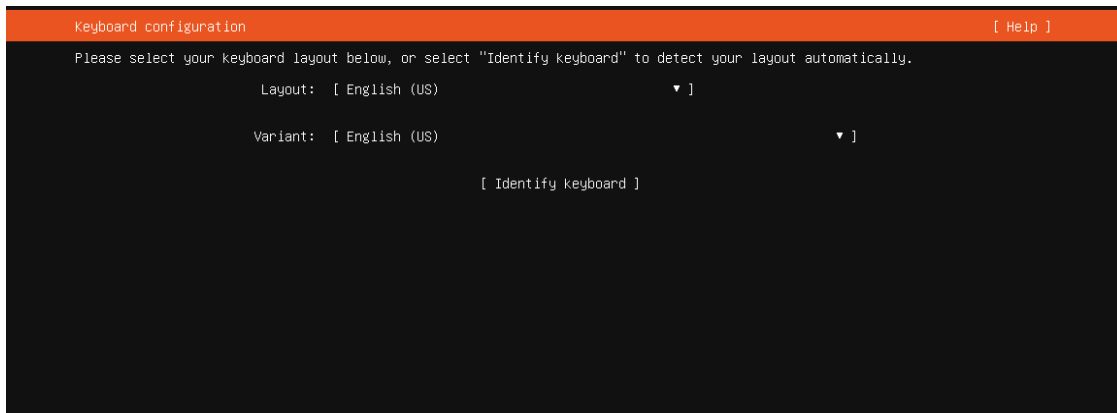
Begin: Setting hostname... .. done.
Begin: Setting up console keyboard... .. done.
Begin: Applying desktop settings... .. done.
Begin: Regenerating SSL certificate... .. done.
Begin: Loading preseed file... .. done.
Begin: Adding live session user... .. passwd: password changed.
done.
Begin: Setting up init... .. done.
Begin: Configuring accessibility options... .. done.
Begin: Disabling update-notifier... .. done.
Begin: Enabling detection of crashes... .. done.
Begin: Disabling unnecessary KDE services... .. done.
Begin: Disabling the lock screen for LXQt (LP: #2000787)... .. done.
Begin: Generating APT cache for cdrom ... Ign:1 file:/cdrom resolute InRelease
Get:2 file:/cdrom resolute Release [664 B]
Get:2 file:/cdrom resolute Release [664 B]
Get:3 file:/cdrom resolute Release.gpg [228 B]
Get:3 file:/cdrom resolute Release.gpg [228 B]
Get:4 file:/cdrom resolute/main amd64 Packages [39.1 kB]
Get:5 file:/cdrom resolute/restricted amd64 Packages [13.9 kB]
done.
Begin: Running /scripts/nfs-bottom ... done.
Begin: Running /scripts/init-bottom ... Begin: Stopping multipathd ... done.
done.
—

```

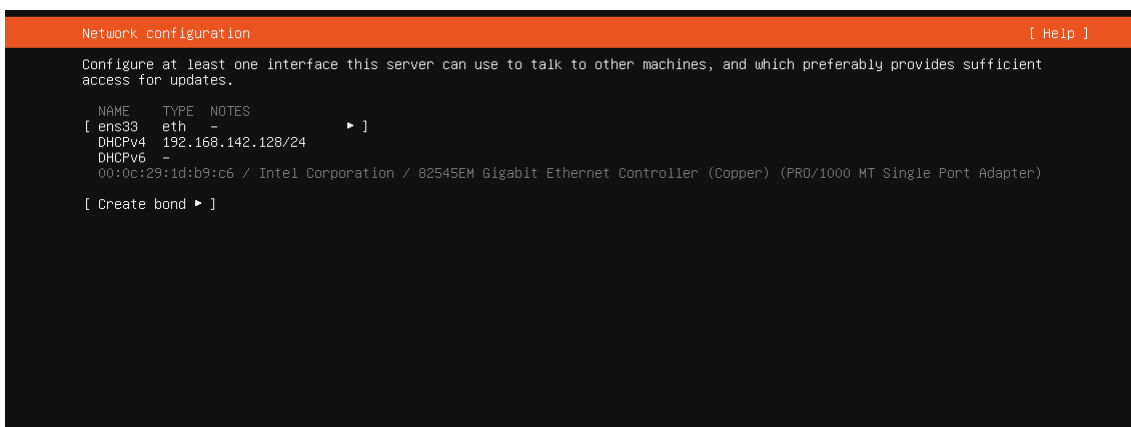
9. **ubuntu_language.png** — Shows English selected as the installation lang



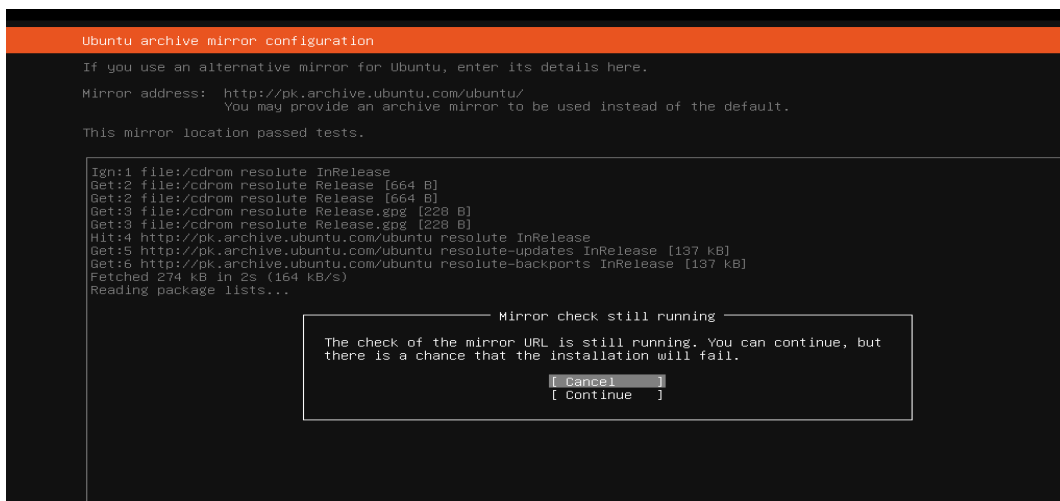
10. **ubuntu_keyboard_layout.png** — Shows the selected keyboard layout.



11. **ubuntu_network_interface.png** — Shows the detected network interface.



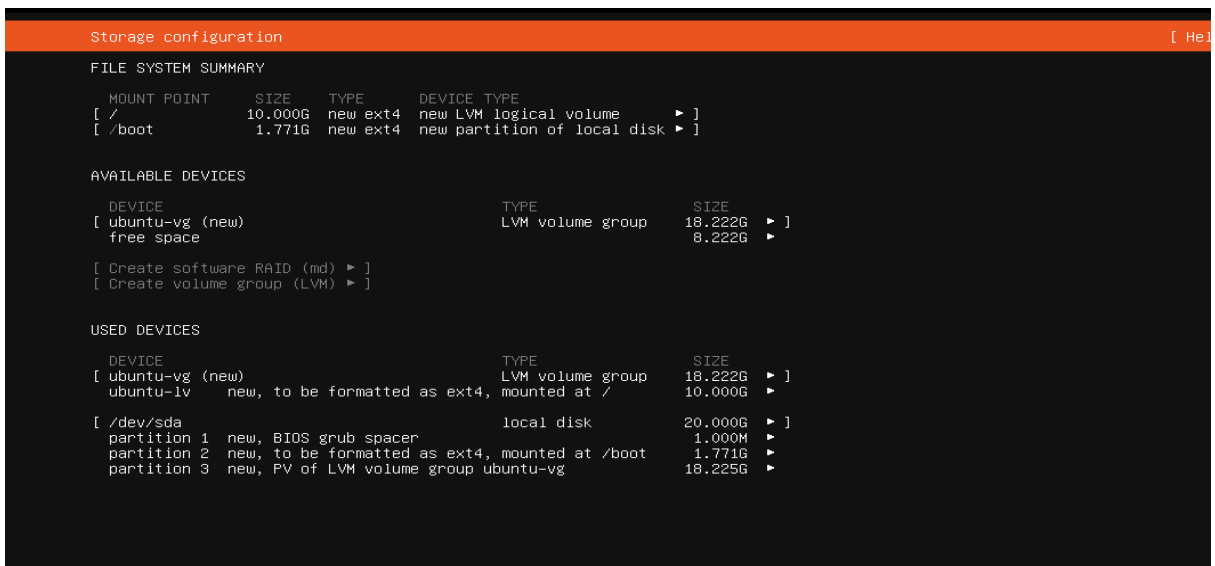
12. **ubuntu_installer_network.png** — Shows the Ubuntu installer network configuration.



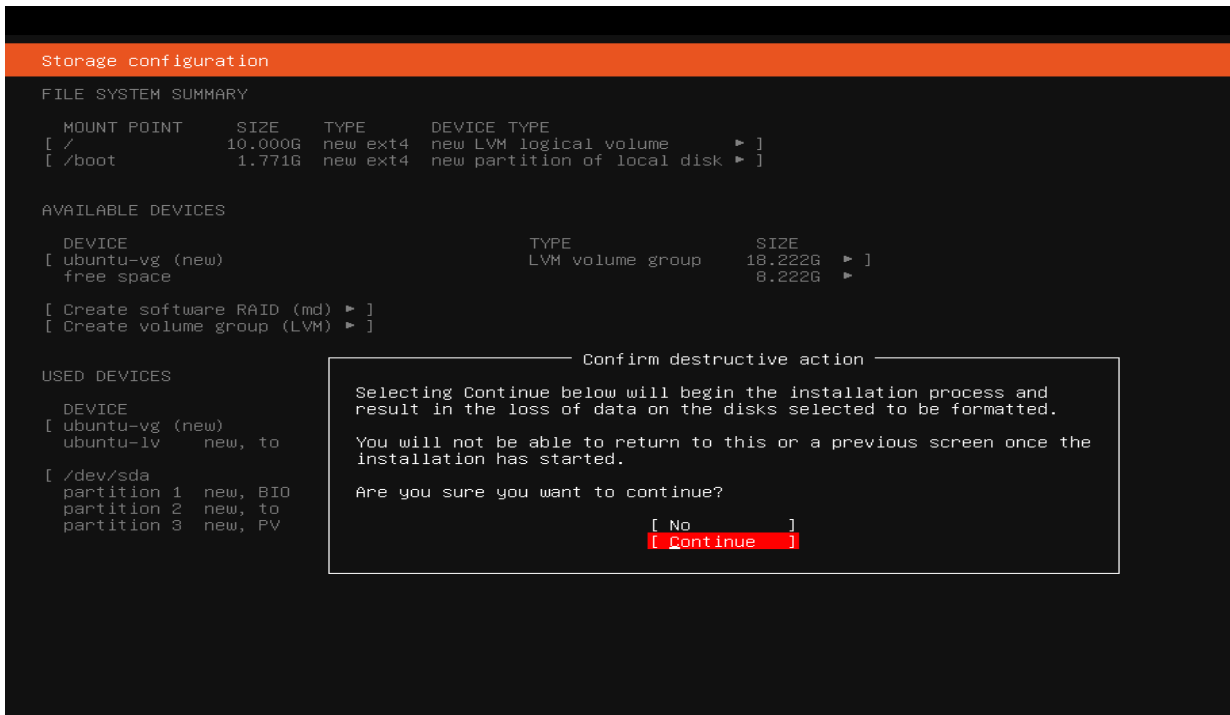
13. **ubuntu_guided_storage.png** — Shows guided storage with the virtual disk selected.



14. **ubuntu_storage_configuration.png** — Shows the configured partitions and LVM storage.



15. **ubuntu_storage_warning.png** — Shows the confirmation warning before applying storage changes.



16. **ubuntu_profile_setup.png** — Shows the completed Ubuntu profile configuration.



17. **ubuntu_installation_progress.png** — Shows Ubuntu Server installation in progress.

```
Installing system [ Help ]

subiquity/2dev/apply_autoinstall_config:
subiquity/4d/apply_autoinstall_config:
subiquity/Late/apply_autoinstall_config:
configuring apt
curtin command in-target
installing system
executing curtin install initial step
executing curtin install partitioning step
curtin command install
  configuring storage
    running 'curtin block-meta simple'
    curtin command block-meta
      removing previous storage devices
    configuring disk: disk-sda
    configuring partition: partition-0
    configuring partition: partition-1
    configuring partition: partition-2
    configuring format: format-0
    configuring lvm_volgroup: lvm_volgroup-0
    configuring lvm_partition: lvm_partition-0
    configuring format: format-1
    configuring mount: mount-1
    configuring mount: mount-0
  executing curtin install extract step
curtin command install
  writing install sources to disk
  running 'curtin extract'
  curtin command extract
    acquiring and extracting image from cp:///tmp/tmpwsaczf0p/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel \

[ View full log ]
```

18. **ubuntu_installation_complete.png** — Shows successful completion of the Ubuntu installation.

```
Installation complete! [ Help ]

  running 'curtin extract'
  curtin command extract
    acquiring and extracting image from cp:///tmp/tmpwsaczf0p/mount
configuring keyboard
curtin command in-target
executing curtin install curthooks step
curtin command install
  configuring installed system
  running 'curtin curthooks'
  curtin command curthooks
    configuring apt configuring apt
    installing missing packages
    configuring iscsi service
    configuring raid (mdadm) service
    configuring NVMe over TCP
    installing kernel
    setting up swap
    apply networking config
    writing etc/fstab
    configuring multipath
    updating packages on target system
    configuring pollinate user-agent on target
    configuring kernel crash dumps settings
    final kernel configuration
    configuring target system bootloader
    installing grub to target devices
    copying metadata from /cdrom
final system configuration
calculating extra packages to install
installing openssh-server
retrieving openssh-server
curtin command system-install
unpacking openssh-server
curtin command system-install
configuring cloud-init
downloading and installing security updates
curtin command in-target
restoring apt configuration
  curtin command in-target
subiquity/Late/run_builtin:
subiquity/Late/run_user_supplied:

[ View full log ]
[ Reboot Now ]
```

Ubuntu Verification

19. **ubuntu_login_screen.png** — Shows the Ubuntu Server login screen.

```
ubuntu login: eman25-wq
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Sat Sep 19 01:05:23 PM UTC 2026

  System load:  0.22               Processes:           252
  Usage of /:   33.8% of 9.75GB    Users logged in:    0
  Memory usage: 7%                IPv4 address for ens33: 192.168.142.129
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

eman25-wq@ubuntu:~$
```

20. **ubuntu_identity_verified.png** — Verifies the username eman25-wq and hostname ubuntu.

```
The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

eman25-wq@ubuntu:~$ whoami
eman25-wq
eman25-wq@ubuntu:~$ hostname
ubuntu
eman25-wq@ubuntu:~$
```

21. **ubuntu_ip_addr_command.png** — Shows the ip addr command used to identify the network address.

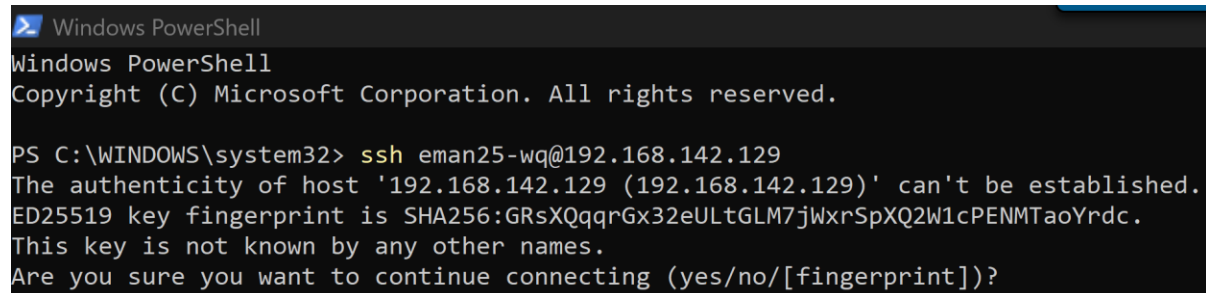
```

ubuntu
eman25-wq@ubuntu:~$ ip addr
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: ens33: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 00:0c:29:1d:b9:c6 brd ff:ff:ff:ff:ff:ff
    altname enp2s1
    altname enx00c291db9c6
    inet 192.168.142.129/24 metric 100 brd 192.168.142.255 scope global dynamic ens33
        valid_lft 1170sec preferred_lft 1170sec
    inet6 fe80::20c:29ff:fe1d:b9c6/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
eman25-wq@ubuntu:~$

```

Windows SSH

22. **windows_ssh_command.png** — Shows the SSH command used to connect from Windows.



```

Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\WINDOWS\system32> ssh eman25-wq@192.168.142.129
The authenticity of host '192.168.142.129 (192.168.142.129)' can't be established.
ED25519 key fingerprint is SHA256:GRsXQqqrGx32eULtGLM7jWxrSpXQ2W1cPENMTaoYrdc.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])?

```

23. **windows_ssh_fingerprint.png** — Shows the SSH host fingerprint confirmation.

```

Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '192.168.142.129' (ED25519) to the list of known hosts.
eman25-wq@192.168.142.129's password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-31-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Sat Sep 19 02:15:32 PM UTC 2026

System load:  0.0               Processes:           219
Usage of /:   33.8% of 9.75GB   Users logged in:    0
Memory usage: 7%               IPv4 address for ens33: 192.168.142.129
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

13 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

eman25-wq@ubuntu:~$

```

PDF

24. **solution_title_page.png** — Shows the title page of the completed lab solution.

CC / Labs / Lab01 / screenshots /

Add file ▾

eman25-wq
 Add files via upload
 0526646 · now
 History

Name	Last commit message	Last commit date
..		
.gitkeep	Create .gitkeep	9 minutes ago
github_profile.png	Add files via upload	now
ubuntu_guided_storage.png.png	Add files via upload	now
ubuntu_installation_complete.png.png	Add files via upload	now
ubuntu_installation_progress.png.png	Add files via upload	now
ubuntu_installer_boot.png.png	Add files via upload	now
ubuntu_installer_network.png.png	Add files via upload	now

25. **lab1_solution_pdf.png** — Shows the completed Lab1_Solution.pdf.

CC / Labs / Lab01 / screenshots / 

 eman25-wq Add files via upload

Name	Last commit message
 ..	
 .gitkeep	Create .gitkeep
 Lab1_Solution.docx	Add files via upload
 github_profile.png	Add files via upload

